

VELOX V5 — Market-Readiness Roadmap

Built on V4 (all 10 phases complete, 2026-05-24). If you are Claude: read **CLAUDE.md** first, then **VELOX_V4_ROADMAP.md §16**, then this file.

V5 Philosophy

V1–V4 built a technically complete product. V5 makes it a **sellable** product.

A phase belongs in V5 if and only if a real prospect would say *"we can't adopt without this."* Everything else is parked in **VELOX_FUTURE.md**.

V5 is organised by **buyer-adoption blockers**, not by engineering aesthetics:

1. **Tier 1 — Self-serve adoption blockers** (V5-0, V5-1, V5-2): without these a developer cannot evaluate Velox at all.
2. **Tier 2 — Enterprise deal blockers** (V5-3, V5-4, V5-5, V5-6): without these every deal above ~\$50k ACV stalls at procurement, security, or finance review.
3. **Tier 3 — Competitive parity & launch surface** (V5-7, V5-8, V5-9): polish that wins bake-offs and converts the funnel.

Rules for every V5 phase:

1. Every phase must answer "which adoption blocker does this remove?" in one sentence.
2. Each phase ships independently — no phase blocks the next on merge.
3. Backend, SDK, and frontend phases follow the V4 separation: do not mix them.
4. Every phase must list at least one external signal that proves it removed the blocker (download count, SSO test login, customer migration, etc.).

Locked decisions (made before V5-0 starts — see §17):

- License: Elastic License v2 (replaces MIT)
 - Brand: independent (no "Anthropic" anywhere)
 - CLI shape: single **velox** binary with subcommands
 - OpenAPI tool: **utoipa** + **utoipa-swagger-ui**
 - SSO order: OIDC first, then SAML — both in V5-3
 - Managed cloud: deferred to V6 (not V5 scope)
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1. Complete Gap Audit

Three lenses: **API surface**, **enterprise readiness**, and **funnel/GTM**.

API surface gaps (blocks self-serve adoption)

Gap	Why it blocks adoption	Phase
No /v1/embeddings endpoint	RAG apps (the largest LLM workload) cannot use Velox at all	V5-0
No /v1/images/generations , /v1/audio/*	Multimodal apps cannot unify on Velox	V5-0
No /v1/models	OpenAI SDK calls <code>client.models.list()</code> on init; many libraries break	V5-0
No /v1/completions (legacy)	Older codebases (LangChain ≤ 0.1 , llama-index ≤ 0.10) cannot point at Velox	V5-0
No tools / tool_calls extracted in audit log	Function-calling apps have no observability	V5-0
No official Python / Node SDKs	Developer evaluation gives up in <15 minutes	V5-1
No OpenAPI spec	No autocomplete, no codegen, no Postman	V5-1
No CLI for ops	Operators expect velox keys create , velox migrate , velox doctor	V5-1
No Helm chart	Enterprises will not paste raw YAML from a README	V5-2
No Terraform provider	IaC users cannot manage Velox declaratively	V5-2
No migration importers (LiteLLM, Portkey, OpenRouter)	Switching cost kills every replacement deal	V5-2

Gap	Why it blocks adoption	Phase
No one-click deploy (Railway/Fly/Render)	Demo-day prospects need a 60-second deploy path	V5-2
No backup/restore CLI	Ops will not run <code>pg_dump</code> by hand against a vendor product	V5-2

Enterprise gaps (blocks deals over ~\$50k ACV)

Gap	Why it blocks adoption	Phase
No SSO / OIDC	Hard blocker for any company with central IAM	V5-3a
No SAML 2.0	Hard blocker for traditional enterprise IT	V5-3b
No SCIM provisioning	Hard blocker for IT-led rollouts	V5-3
Audit log not tamper-evident	Fails SOC 2 CC7.2 controls	V5-4
No data export (DSAR / GDPR Article 15)	EU prospects walk during legal review	V5-4
No right-to-be-forgotten	GDPR Article 17 blocker	V5-4
No secrets manager integration (Vault, AWS SM, GCP SM)	Production teams do not use env vars for secrets	V5-4
No data residency controls	Cannot answer "stays in EU?"	V5-4
No cost attribution by team/project/user	Finance teams need chargeback splits	V5-5
No budget forecasting	Cannot answer "when will we hit cap?"	V5-5
No invoice generation / chargeback CSV	Finance builds the spreadsheets externally	V5-5
No Slack / PagerDuty / Email / Teams channels	Webhook-only feels half-finished	V5-6
No Datadog / New Relic exporters	"We can't see Velox in our APM" disqualifies	V5-6
No SLA reports per workspace	Mid-market customers ask for monthly proof	V5-6
No admin-configurable policy engine	Security teams ask "what stops a dev sending PII to OpenAI?"	V5-7
No prompt injection detection	AI-security teams ask	V5-7
No content moderation hook	Required by regulated industries	V5-7

Funnel / GTM gaps (blocks ever being considered)

Gap	Why it blocks adoption	Phase
No public load test numbers	"How many req/s?" — no answer means lost bake-off	V5-8

Gap	Why it blocks adoption	Phase
No chaos test report	Reliability claims read as marketing without proof	V5-8
No E2E Playwright tests	Quality signal; V5 regressions otherwise certain	V5-8
No marketing landing page	Visitors leave without conversion	V5-9
No comparison pages (vs LiteLLM, Portkey, Helicone, Cloudflare)	You do not appear in consideration sets	V5-9
No interactive product tour	Conversion drop between sign-up and first request	V5-9
No sample apps repo (RAG, agent, embeddings search)	Devs cannot picture using Velox in their stack	V5-9
No public Discord / community	No defensible adoption surface	V5-9
No status page	Operators check before adoption — empty = no	V5-9
MIT license open to AWS-style commodification	Future managed offering unprotected	V5-9 (license switch ships with launch)
README still attributes to Anthropic	Brand confusion blocks credibility for solo founder	V5-9

2. V5 Testing Philosophy

Inherits V2/V3/V4 regression contract. Every phase gate-in and gate-out:

```
cargo test
cargo clippy -- -D warnings
cargo fmt -- --check
```

V5 backend tests live in `tests/v5/`. Frontend phases have no Rust tests (they are tested via Playwright in V5-8 and visually during the phase).

```
tests/
├── v2/    ← must stay green, never touch
├── v3/    ← must stay green, never touch
├── v4/    ← must stay green, never touch
└── v5/
    ├── common.rs
    ├── v5_0_api_expansion.rs
    ├── v5_1_openapi.rs
    └── v5_2_migration_imports.rs
```

- v5_3_oidc.rs
- v5_3_saml.rs
- v5_3_scim.rs
- v5_4_audit_chain.rs
- v5_4_data_export.rs
- v5_4_secrets_manager.rs
- v5_5_cost_attribution.rs
- v5_5_budget_forecast.rs
- v5_6_notifications.rs
- v5_6_apm_exporters.rs
- v5_7_policy_engine.rs

Test naming: `v5_{phase}_{feature}_{expected_outcome}` — underscore, never the letter `p` (CLAUDE.md §11).

External signals per phase (proof the blocker was removed):

Phase	External signal
V5-0	RAG sample app produces <code>/v1/embeddings</code> → semantic search loop end-to-end
V5-1	<code>pip install velox</code> + 3-line example posts to <code>/v1/chat/completions</code>
V5-2	<code>velox import litellm config.yaml</code> converts a real LiteLLM config to working Velox state
V5-3	Test login via Auth0 + Google Workspace; SCIM provisioning from Okta sandbox
V5-4	<code>velox audit verify</code> passes on 100k-row chain; export → re-import roundtrip lossless
V5-5	Cost report split by 3 tag dimensions matches <code>sum(requests.cost_usd)</code> exactly
V5-6	Test alert fires into a real Slack workspace + PagerDuty trigger
V5-7	Policy rule blocks a PII-containing request in dry-run mode with audit trail
V5-8	k6 + criterion CI runs produce a public benchmark page; chaos report committed
V5-9	Landing site live at <code>velox.dev</code> ; license switched to ELv2; "Anthropic" removed

3. Phase V5-0: API Surface Expansion

Goal: Make Velox usable for every real LLM workload, not just chat completions.

3.1 Problem

The OpenAI-compatible surface today is only `POST /v1/chat/completions`. Any application that calls `client.embeddings.create()`, `client.models.list()`, `client.images.generate()`, or `client.audio.transcriptions.create()` cannot use Velox without changing application code. The OpenAI SDK calls `models.list()` on construction for many integrations — so even chat-only apps fail to initialise.

3.2 Endpoints to add

Endpoint	Cache	Cost tracked	Stream	Notes
POST /v1/embeddings	Exact only (semantic doesn't apply to itself)	Yes	No	Largest gap — RAG depends on this
GET /v1/models	No (5s in-memory TTL)	No	No	Aggregated from enabled providers
POST /v1/images/generations	No (high cost, low repetition)	Yes	No	Pricing per image, not per token
POST /v1/audio/transcriptions	No (binary input)	Yes (per second)	No	multipart/form-data passthrough
POST /v1/audio/speech	Exact (same text → same audio)	Yes (per char)	Yes (audio chunks)	Binary streaming response
POST /v1/completions (legacy)	Yes (both layers)	Yes	Yes	Wraps as chat internally — minimal extra code

3.3 Function-calling audit

Extract **tools** from request body and **tool_calls** from response body. Store as JSONB on **requests** table for analytics queries:

```
-- migrations/0027_api_expansion.sql
ALTER TABLE requests
  ADD COLUMN tool_calls JSONB,
  ADD COLUMN endpoint VARCHAR(50) NOT NULL DEFAULT 'chat/completions';
CREATE INDEX idx_requests_endpoint ON requests(endpoint);
CREATE INDEX idx_requests_tool_calls_gin ON requests USING gin(tool_calls);

-- Extend model_pricing for non-text modalities
ALTER TABLE model_pricing
  ADD COLUMN price_per_image DECIMAL(12,8),
  ADD COLUMN price_per_audio_second DECIMAL(12,8),
  ADD COLUMN price_per_character DECIMAL(12,8);
```

3.4 Provider trait extensions

Add to **src/providers/mod.rs**:

```
#[async_trait]
pub trait Provider: Send + Sync {
```

```

    // existing
    async fn chat_completion(...);
    async fn chat_completion_stream(...);
    // new in V5-0
    async fn embeddings(&self, req: EmbeddingsRequest) ->
Result<EmbeddingsResponse>;
    async fn list_models(&self) -> Result<Vec<ModelInfo>>;
    async fn images_generate(&self, req: ImagesRequest) ->
Result<ImagesResponse> {
    Err(ProviderError::Unsupported) // default - opt-in per provider
}
    async fn audio_transcribe(&self, req: TranscribeRequest) ->
Result<TranscribeResponse> {
    Err(ProviderError::Unsupported)
}
    async fn audio_speech(&self, req: SpeechRequest) ->
Result<SpeechStream> {
    Err(ProviderError::Unsupported)
}
}

```

Default impls return **Unsupported** so existing adapters (Bedrock, DeepSeek) don't have to implement every modality before V5-0 ships.

3.5 Files to create

- `src/handlers/gateway/embeddings.rs`
- `src/handlers/gateway/models.rs`
- `src/handlers/gateway/images.rs`
- `src/handlers/gateway/audio.rs`
- `src/handlers/gateway/completions_legacy.rs`
- `src/providers/embeddings.rs` — shared request/response shapes
- `src/gateway/tool_extract.rs` — extracts **tools/tool_calls** for audit log

3.6 Files to modify

- `src/providers/openai.rs` — implement all 5 new trait methods
- `src/providers/anthropic.rs` — embeddings, models (no images/audio)
- `src/providers/gemini.rs`, `groq.rs`, `deepseek.rs` — embeddings + models where supported
- `src/providers/bedrock.rs` — embeddings (Titan/Cohere) + models
- `src/gateway/pipeline.rs` — extract **tool_calls** into request record
- `src/db/requests.rs` — **insert_request** accepts **tool_calls** + **endpoint**
- `src/pricing/mod.rs` — handle image / audio / character pricing
- `src/routes/mod.rs` — wire new routes

3.7 Test contract

File: `tests/v5/v5_0_api_expansion.rs`

```

async fn v5_0_embeddings_endpoint_returns_openai_shape()
async fn v5_0_embeddings_exact_cache_hit_returns_cached()
async fn v5_0_embeddings_cost_tracked_in_requests()
async fn v5_0_embeddings_routes_via_priority()
async fn v5_0_list_models_aggregates_across_providers()
async fn v5_0_list_models_cached_for_5_seconds()
async fn v5_0_images_endpoint_passes_through()
async fn v5_0_images_cost_uses_price_per_image()
async fn v5_0_audio_transcription_multipart_upload_works()
async fn v5_0_audio_speech_streams_chunks()
async fn v5_0_completions_legacy_proxies_to_chat()
async fn v5_0_tool_calls_extracted_into_requests_row()
async fn v5_0_endpoint_field_set_per_route()
async fn v5_0_unsupported_modality_returns_404_with_provider_hint()
async fn v5_0_regression_chat_completions_unaffected()

```

3.8 Definition of done

```

cargo test v5_0
cargo test
cargo clippy -- -D warnings

# External signal: sample RAG app at examples/rag-chatbot/ uses
# /v1/embeddings → pgvector → /v1/chat/completions in one process
cd examples/rag-chatbot && python main.py "what is velox?"

```

4. Phase V5-1: SDKs, CLI, and OpenAPI

Goal: Make integration a single-line code change for any developer, and let operators administer Velox without `curl`.

4.1 OpenAPI spec generation

Adopt `utoipa` + `utoipa-swagger-ui`. Every existing admin handler gets a `#[utoipa::path(...)]` annotation. Spec is generated at compile time, served at:

- `GET /admin/openapi.json` — machine-readable OpenAPI 3.1
- `GET /admin/docs` — Swagger UI (no JS bundling — `utoipa-swagger-ui` ships static assets)

```

# Cargo.toml - V5-1
utoipa = { version = "5", features = ["axum_extras", "uuid", "chrono",
"decimal"] }
utoipa-swagger-ui = { version = "8", features = ["axum"] }

```

4.2 `velox` CLI

Single binary, **clap**-based, subcommand structure:

```

velox serve                                # runs the server (default if no
subcommand)
velox doctor                               # readiness checks (existing --doctor
logic)
velox demo                                # demo mode (existing --demo logic)

velox keys list
velox keys create --name X --budget 100 --rpm 60
velox keys rotate <id>
velox keys revoke <id>

velox migrate up
velox migrate down
velox migrate status

velox config get
velox config set log_request_bodies=true

velox import litellm <config.yaml>
velox import portkey <export.json>
velox import openrouter

velox replay <request-id>
velox audit verify                         # (lands fully in V5-4)
velox backup <dest>                       # (lands fully in V5-2)
velox restore <src>                       # (lands fully in V5-2)

```

The CLI talks to a running Velox via the admin API using a **velox.cli.toml** file (**url**, **admin_token**). When run on the same host as the server it can also speak directly to the DB for **migrate/backup/restore**.

4.3 Python SDK (**velox-python**, separate repo)

Thin wrapper around the official **openai** Python SDK so users get full LLM SDK ergonomics + Velox-specific extras:

```

from velox import Velox

client = Velox(url="https://velox.acme.com", api_key="vx-sk-...")

# OpenAI-compatible (just like openai.OpenAI)
client.chat.completions.create(model="gpt-4o", messages=[...])

# Velox-specific helpers
client.keys.create(name="prod", budget=100)
client.analytics.overview(period="30d")
client.replay("req-uuid")
client.cache.flush()

```

```
# Response headers exposed as typed attributes
resp = client.chat.completions.create(...)
resp.velox.cache_hit           # "exact" | "semantic" | None
resp.velox.cache_similarity    # 0.94 | None
resp.velox.downgraded          # "cost_optimized" | None
resp.velox.audit_hash          # str
```

Auto-generated from the OpenAPI spec for `client.keys/analytics/replay/cache` methods. Hand-coded thin layer over `openai` SDK for chat/embeddings/etc.

4.4 Node.js / TypeScript SDK (`velox-node`, separate repo)

Same surface in TypeScript, wraps `openai` npm package. Published to npm as `velox`.

4.5 Postman collection

Generated from OpenAPI via `openapi-to-postmanv2`. Shipped at `docs/postman/velox.postman_collection.json`. CI regenerates on every release.

4.6 Files to create (server side)

- `src/cli/mod.rs` — clap parser
- `src/cli/keys.rs`, `migrate.rs`, `config.rs`, `import.rs`, `backup.rs`
- `src/openapi.rs` — `OpenApiDoc` derive root combining all handler annotations
- `src/handlers/admin/docs.rs` — serves Swagger UI + JSON

4.7 Files to modify

- `src/main.rs` — replace ad-hoc `--doctor` / `--demo` flag parsing with `velox::cli::run()`
- Every `src/handlers/admin/*.rs` — add `#[utoipa::path(...)]` annotations
- `Cargo.toml` — add `utoipa`, `utoipa-swagger-ui`, `clap` features (`derive`, `env`)

4.8 Test contract

File: `tests/v5/v5_1_openapi.rs`

```
async fn v5_1_openapi_json_endpoint_returns_valid_spec()
async fn v5_1_openapi_includes_all_admin_endpoints()
async fn v5_1_openapi_spec_validates_against_3_1_schema()
async fn v5_1_swagger_ui_endpoint_returns_200()
fn v5_1_cli_keys_create_invokes_admin_api()
fn v5_1_cli_migrate_status_reads_migrations_table()
fn v5_1_cli_config_get_reads_admin_config()
fn v5_1_cli_help_lists_all_subcommands()
async fn v5_1_regression_existing_handler_responses_unchanged()
```

SDK tests live in their own repos (`velox-python/tests/`, `velox-node/test/`) and are gated by their own CI.

4.9 Definition of done

```
cargo test v5_1
cargo test
cargo clippy -- -D warnings

# External signal: 3-line Python script works
pip install velox
python -c "from velox import Velox; v=Velox(url='http://localhost:8080',
api_key='vx-sk-...'); print(v.chat.completions.create(model='gpt-4o-mini',
messages=[{'role':'user','content':'hi'}]).choices[0].message.content)"
```

5. Phase V5-2: Deployment & Migration Tooling

Goal: Eliminate the deploy and switching-cost friction that keeps prospects on competitors.

5.1 Helm chart

`charts/velox/` containing:

- `values.yaml` with sensible defaults
- HPA (CPU + custom metric: requests/sec)
- ServiceMonitor for Prometheus Operator
- Ingress (nginx/traefik examples)
- Secrets references for `JWT_SECRET`, `ENCRYPTION_KEY`, provider keys
- External Postgres reference (does not bundle Postgres — see VELOX_FUTURE §3)
- Optional Qdrant subchart toggle

Published to a GitHub Pages helm repo (<https://farzad.github.io/velox-charts>).

5.2 Terraform provider (`terraform-provider-velox`, separate repo)

Resources:

- `velox_provider`
- `velox_api_key`
- `velox_workspace`
- `velox_alert`
- `velox_prompt`
- `velox_workspace_member`

Built against the OpenAPI spec from V5-1 via `terraform-plugin-framework`. Published to the Terraform Registry.

5.3 One-click deploy

Platform	Artifact	Path
Railway	<code>railway.json</code> + Postgres template	<code>deploy/railway/</code>
Fly.io	<code>fly.toml</code> + <code>fly launch</code> config	<code>deploy/fly/</code>
Render	<code>render.yaml</code> Blueprint	<code>deploy/render/</code>
AWS Marketplace AMI	Packer config + launch script	<code>deploy/aws-ami/</code> (defer to V6 if time-constrained)

5.4 Migration importers (CLI commands from V5-1)

Each importer reads the competitor's config format and creates equivalent Velox state via the admin API:

```
velox import litellm    path/to/proxy_config.yaml
velox import portkey    path/to/portkey_export.json
velox import openrouter # uses OpenRouter's public model API
```

LiteLLM mapping (reads `model_list:` and `general_settings:`):

- LiteLLM model entry → Velox provider + key
- LiteLLM `router_settings.routing_strategy` → Velox `routing_strategy` on `api_keys`
- LiteLLM `litellm_settings.cache` → Velox cache config

Portkey mapping:

- Virtual key → Velox `api_key`
- Provider config → Velox provider
- Routing rules → Velox `routing_strategy` (best-effort; complex rules become V5-7 policies)

5.5 Backup / Restore CLI

```
velox backup ./velox-backup-2026-06-01.tar.gz
# Bundles:
#   - pg_dump of velox DB
#   - models/ directory (ONNX + tokenizer)
#   - velox.toml
#   - schema version stamp

velox restore ./velox-backup-2026-06-01.tar.gz
# Restores all of the above, with version compatibility check
```

Wraps `pg_dump/pg_restore` under the hood, but provides a single artifact and version safety. Cron example documented.

5.6 HA deployment guide

New doc: [docs/deployment/ha.md](#):

- Postgres primary + replica setup
- ≥ 2 Velox nodes behind LB (V2-6 clustering already supports this)
- External Qdrant (V4-9)
- Encryption key rotation procedure
- DR runbook

5.7 Files to create

- [charts/velox/](#) — full Helm chart
- [deploy/railway/railway.json](#), [deploy/fly/fly.toml](#), [deploy/render/render.yaml](#)
- [src/cli/import/litellm.rs](#), [portkey.rs](#), [openrouter.rs](#)
- [src/cli/backup.rs](#)
- [docs/deployment/ha.md](#)
- [docs/deployment/helm.md](#)
- [docs/deployment/terraform.md](#)

5.8 Test contract

File: [tests/v5/v5_2_migration_imports.rs](#)

```
fn v5_2_litellm_yaml_parsing_to_provider_list()
fn v5_2_litellm_routing_strategy_maps_correctly()
fn v5_2_portkey_export_parsing_to_provider_list()
fn v5_2_openrouter_import_creates_model_aliases()
fn v5_2_backup_produces_complete_archive()
fn v5_2_restore_roundtrip_preserves_all_tables()
fn v5_2_restore_rejects_incompatible_version()
fn v5_2_helm_chart_lint_passes()    # invokes `helm lint charts/velox`
fn v5_2_helm_template_produces_valid_k8s_yaml()
```

5.9 Definition of done

```
cargo test v5_2
helm lint charts/velox
helm template charts/velox | kubeval

# External signal: real LiteLLM proxy_config.yaml converts cleanly
velox import litellm test-fixtures/litellm-sample.yaml
# → reports N providers created, K keys created, 0 errors
```

6. Phase V5-3: Enterprise Auth (OIDC, SAML, SCIM)

Goal: Unblock every deal where IT controls login.

This is a large phase. It ships in three sub-phases, each independently shippable.

6.1 Schema

```
-- migrations/0028_sso_identity.sql
CREATE TABLE identity_providers (
  id                UUID PRIMARY KEY,
  workspace_id      UUID NOT NULL REFERENCES workspaces(id) ON DELETE
CASCADE,
  kind              VARCHAR(20) NOT NULL,      -- 'oidc' | 'saml'
  name              VARCHAR(100) NOT NULL,
  config            JSONB NOT NULL,             -- discovery URL / IdP
  metadata / client_id etc.
  group_role_map    JSONB NOT NULL DEFAULT '{}', -- {"velox-
admin":"admin",...}
  enabled           BOOLEAN NOT NULL DEFAULT TRUE,
  created_at        TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE TABLE identities (
  id                UUID PRIMARY KEY,
  user_id           UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  idp_id            UUID NOT NULL REFERENCES identity_providers(id) ON DELETE
CASCADE,
  external_id       TEXT NOT NULL,
  last_login        TIMESTAMPTZ,
  UNIQUE (idp_id, external_id)
);

CREATE TABLE scim_tokens (
  id                UUID PRIMARY KEY,
  workspace_id      UUID NOT NULL REFERENCES workspaces(id) ON DELETE CASCADE,
  name              VARCHAR(100) NOT NULL,
  token_hash        TEXT NOT NULL UNIQUE,
  created_at        TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  revoked_at        TIMESTAMPTZ
);
```

6.2 Sub-phase V5-3a — OIDC + SCIM

OIDC:

- `openidconnect` crate handles Authorization Code + PKCE flow
- New routes: `GET /auth/oidc/:idp_id/start`, `GET /auth/oidc/:idp_id/callback`
- On callback: verify ID token, look up `identities` row, JIT-create user if absent, mint Velox JWT
- Group claims → Velox role via `group_role_map`

SCIM 2.0:

- New routes under `/scim/v2/`:
 - `GET/POST /Users`

- GET/PUT/PATCH/DELETE /Users/:id
- GET/POST /Groups
- Auth via **Authorization: Bearer <scim_token>** (separate from admin JWT)
- Idempotent operations; mapped to existing **users** + **workspace_members** tables

Dashboard:

- **/settings/sso** page — connect IdP, paste discovery URL, configure group mapping, "test connection"
- **/settings/scim** page — generate SCIM token, copy once, list active integrations

6.3 Sub-phase V5-3b — SAML 2.0

- **samael** crate for SAML assertion validation
- New routes: **POST /auth/saml/:idp_id/acs**, **GET /auth/saml/:idp_id/metadata**
- IdP-initiated and SP-initiated flows
- Sign-out via SAML SLO (best-effort)
- Same **identities** table; **idp_kind = 'saml'**

6.4 Files to create

- **src/auth/oidc.rs**
- **src/auth/saml.rs**
- **src/auth/scim.rs**
- **src/handlers/auth/sso.rs**
- **src/handlers/scim/users.rs, groups.rs**
- **src/db/identities.rs**
- **dashboard/src/app/(dashboard)/settings/sso/page.tsx**
- **dashboard/src/app/(dashboard)/settings/scim/page.tsx**

6.5 Test contract

File: **tests/v5/v5_3_oidc.rs**

```
async fn v5_3_oidc_callback_creates_user_jwt()
async fn v5_3_oidc_callback_returns_existing_user_on_subsequent_login()
async fn v5_3_oidc_group_claim_maps_to_role()
async fn v5_3_oidc_invalid_id_token_rejected()
async fn v5_3_oidc_state_param_csrf_protected()
async fn v5_3_oidc_disabled_idp_returns_404()
```

File: **tests/v5/v5_3_saml.rs**

```
async fn v5_3_saml_acs_valid_assertion_creates_session()
async fn v5_3_saml_invalid_signature_rejected()
async fn v5_3_saml_expired_assertion_rejected()
```

```
async fn v5_3_saml_metadata_endpoint_returns_xml()
async fn v5_3_saml_group_attribute_maps_to_role()
```

File: `tests/v5/v5_3_scim.rs`

```
async fn v5_3_scim_create_user_succeeds_with_token()
async fn v5_3_scim_create_user_rejects_without_token()
async fn v5_3_scim_update_user_deactivation_revokes_access()
async fn v5_3_scim_get_users_paginated()
async fn v5_3_scim_delete_user_cascades_to_memberships()
async fn v5_3_scim_token_revocation_blocks_further_calls()
```

6.6 Definition of done

```
cargo test v5_3
cargo test
cargo clippy -- -D warnings

# External signals:
# - Successful OIDC login from Google Workspace test tenant
# - Successful SAML login from Auth0 test tenant
# - SCIM provisioning from Okta sandbox creates 5 users + revokes 1
```

7. Phase V5-4: Compliance & Audit Hardening

Goal: Pass a SOC 2 Type II readiness review and a GDPR/HIPAA RFP without ad-hoc changes.

7.1 Tamper-evident audit chain

Each row in `requests` carries the SHA-256 of the previous row's audit hash, forming a chain.

```
-- migrations/0029_audit_chain.sql
ALTER TABLE requests
  ADD COLUMN audit_hash      TEXT,
  ADD COLUMN previous_hash  TEXT,
  ADD COLUMN chain_sequence BIGSERIAL;
CREATE INDEX idx_requests_chain_sequence ON requests(chain_sequence);
```

Hash inputs: `request_id || api_key_id || created_at || status || cost_usd || tokens_total || previous_hash`.

`velox audit verify [--from <id>] [--to <id>]` walks the chain and reports first mismatch (if any). Existing rows pre-V5-4 are stamped with `chain_sequence = 0` and excluded from verification.

7.2 Data export (DSAR / Article 15)

Streaming JSONL or CSV export of all data tied to a workspace or a single user:

- `GET /admin/export/workspace/:id?format=jsonl` — all requests, keys, alerts for a workspace
- `GET /admin/export/user/:id?format=jsonl` — all DSAR-relevant fields for a single user (their requests across workspaces, account metadata)

Both are admin-only. Streamed via `axum::body::Body` to avoid loading everything into memory. Includes a manifest with row counts and the audit chain range covered.

7.3 Right to be forgotten (Article 17)

```
DELETE /admin/users/:id?cascade=true&redact_audit=true
```

- Removes user, identities, workspace_members rows
- Optionally redacts `request_body.messages[].content` and `response_body.choices[].message.content` to `[redacted]` while preserving the audit row (so the chain stays intact)
- Logs the deletion event itself to `audit_events` table (new) for compliance proof

7.4 Secrets manager integration

Provider keys + encryption keys can be sourced from external secret backends:

```
# velox.toml
encryption_key = "vault://secret/data/velox#encryption_key"
[providers.openai]
api_key = "aws-sm://prod/velox/openai_key"
```

Supported backends:

- `env://VAR_NAME` (default — current behaviour)
- `file:///etc/velox/secrets/openai.key`
- `vault://path/to/secret#field` (HashiCorp Vault, KV v2)
- `aws-sm://secret-name` (AWS Secrets Manager)
- `gcp-sm://projects/p/secrets/s/versions/v` (GCP Secret Manager)
- `k8s://namespace/secret-name#key` (Kubernetes Secret)

Resolved at startup; refresh on SIGHUP for rotation without restart.

7.5 Data residency

```
-- migrations/0030_workspace_region.sql
ALTER TABLE workspaces
```

```

    ADD COLUMN region VARCHAR(20) DEFAULT 'global',
    ADD COLUMN allowed_regions TEXT[];
ALTER TABLE providers
    ADD COLUMN region VARCHAR(20);

```

Router enforces: a workspace with `region = 'eu'` can only select providers with `region IN ('eu', NULL)` (NULL = unrestricted). Violation → 403 with explanation header. Configurable per-workspace via `/settings/region`.

7.6 Compliance docs

New section `docs/compliance/`:

- `soc2.md` — control matrix mapping SOC 2 Trust Service Criteria → Velox features
- `gdpr.md` — Articles 15, 17, 20, 32 → Velox features
- `hipaa.md` — BAA-readiness checklist (encryption, audit, access controls)
- `cloud-acts.md` — what data leaves the user's infrastructure (none; clarify "self-hosted means self-hosted")

7.7 Files to create

- `src/audit/chain.rs`
- `src/handlers/admin/export.rs`
- `src/handlers/admin/users_delete.rs`
- `src/secrets/mod.rs` — Backend trait
- `src/secrets/env.rs`, `file.rs`, `vault.rs`, `aws_sm.rs`, `gcp_sm.rs`, `k8s.rs`
- `src/cli/audit.rs` — velox audit verify
- `docs/compliance/soc2.md`, `gdpr.md`, `hipaa.md`, `cloud-acts.md`

7.8 Test contract

File: `tests/v5/v5_4_audit_chain.rs`

```

async fn v5_4_first_row_after_migration_has_genesis_previous_hash()
async fn v5_4_each_row_chains_correctly()
async fn v5_4_verify_passes_on_intact_chain()
async fn v5_4_verify_detects_tampered_row()
async fn v5_4_verify_detects_deleted_row()
async fn v5_4_chain_sequence_monotonic_under_concurrency()

```

File: `tests/v5/v5_4_data_export.rs`

```

async fn v5_4_workspace_export_streams_all_requests()
async fn v5_4_user_export_includes_cross_workspace_data()
async fn v5_4_export_manifest_row_counts_match_actual()
async fn v5_4_user_delete_cascades_correctly()

```

```
async fn v5_4_user_delete_with_redact_preserves_chain()
async fn v5_4_user_delete_logged_to_audit_events()
```

File: `tests/v5/v5_4_secrets_manager.rs`

```
fn v5_4_env_backend_resolves_var()
fn v5_4_file_backend_resolves_path()
async fn v5_4_vault_backend_resolves_kv2()
async fn v5_4_aws_sm_backend_resolves_secret()
fn v5_4_unknown_scheme_returns_error()
async fn v5_4_sighup_refreshes_secrets()
async fn v5_4_workspace_region_blocks_cross_region_provider()
```

7.9 Definition of done

```
cargo test v5_4
velox audit verify --from beginning # passes on production-shaped data
# External signal: workspace export → re-import roundtrip on a fresh DB →
row counts match
```

8. Phase V5-5: Cost Attribution & FinOps

Goal: Let finance teams answer "who spent what?" without building a side spreadsheet.

8.1 Request tags

Clients send tags via either:

- Header: `X-Velox-Tags: team=acme,project=rag,user=u_123`
- OpenAI-compatible `metadata` field in request body (preferred — already in OpenAI spec)

Stored as JSONB on `requests`:

```
-- migrations/0031_request_tags.sql
ALTER TABLE requests ADD COLUMN tags JSONB DEFAULT '{}';
CREATE INDEX idx_requests_tags_gin ON requests USING gin(tags);
```

8.2 Cost breakdown endpoints

```
GET /admin/analytics/cost?group_by=tag.team&period=30d
GET /admin/analytics/cost?group_by=tag.project&period=30d
GET /admin/analytics/cost?group_by=tag.team,tag.project&period=30d
GET /admin/analytics/cost?group_by=workspace,model&period=30d
```

Returns:

```
{
  "data": {
    "period": "30d",
    "total_cost_usd": 1420.50,
    "groups": [
      {"key": {"tag.team": "acme"}, "cost_usd": 820.40, "request_count": 12400},
      {"key": {"tag.team": "beta"}, "cost_usd": 600.10, "request_count": 8200}
    ]
  }
}
```

8.3 Budget forecasting

Background task every hour computes linear regression on the last 30 days of daily spend per key/workspace. Result stored in `daily_costs` extension:

```
ALTER TABLE daily_costs
  ADD COLUMN forecast_eom_usd DECIMAL(12,8), -- end-of-month projection
  ADD COLUMN forecast_eob_days INTEGER;      -- days until budget
exhausted
```

Dashboard widget on `/overview`: spend trajectory line with budget line overlay; "at current rate you will hit budget on May 18".

Alerts integration: new alert type `forecast_exceeds_budget` triggers when projection crosses the line.

8.4 Invoice generation

```
GET /admin/billing/invoice?workspace=:id&month=2026-04&format=pdf|html|csv
```

Returns:

- PDF (server-side via `printpdf` crate, no headless browser)
- HTML (renderable in dashboard for preview)
- CSV (Stripe-compatible columns for finance imports)

Invoice includes: line items by tag dimension chosen at config time, subtotal, optional markup, total in USD with daily exchange-rate snapshot if needed.

8.5 Chargeback configuration

Workspace-level config in dashboard:

- Primary tag dimension (e.g. **team**)
- Secondary dimension (optional, e.g. **project**)
- Markup percentage (for internal cost recovery)
- Default tag value if missing

8.6 Files to create

- `src/handlers/admin/cost_breakdown.rs`
- `src/handlers/admin/invoice.rs`
- `src/analytics/forecast.rs` — hourly task
- `src/billing/pdf.rs, csv.rs`
- `dashboard/src/app/(dashboard)/billing/page.tsx`
- `dashboard/src/app/(dashboard)/analytics/cost/page.tsx`

8.7 Files to modify

- `src/gateway/pipeline.rs` — extract tags from header or **metadata** body field
- `src/db/requests.rs` — store **tags**
- `src/main.rs` — start forecast background task

8.8 Test contract

File: `tests/v5/v5_5_cost_attribution.rs`

```
async fn v5_5_header_tags_extracted_into_requests()  
async fn v5_5_metadata_body_tags_extracted_into_requests()  
async fn v5_5_header_overrides_body_when_both_present()  
async fn v5_5_cost_breakdown_sums_match_raw_cost()  
async fn v5_5_cost_breakdown_multi_dimension_groups_correctly()  
async fn v5_5_invoice_pdf_renders_without_error()  
async fn v5_5_invoice_csv_matches_stripe_format()  
async fn v5_5_default_tag_value_applied_when_missing()
```

File: `tests/v5/v5_5_budget_forecast.rs`

```
async fn v5_5_forecast_linear_regression_correct_on_known_data()  
async fn v5_5_forecast_handles_sparse_data_without_crash()  
async fn v5_5_forecast_alert_triggers_on_threshold_cross()  
async fn v5_5_forecast_updated_hourly_by_background_task()
```

8.9 Definition of done

```
cargo test v5_5
# External signal: 3-dimension cost breakdown sums match SUM(cost_usd) to
within 1 cent
```

9. Phase V5-6: Notifications & APM Integrations

Goal: Plug into the tools customer oncall and observability teams already use.

9.1 Notification channels (extend V2-2)

New schema:

```
-- migrations/0032_notification_channels.sql
CREATE TABLE notification_channels (
  id          UUID PRIMARY KEY,
  workspace_id UUID REFERENCES workspaces(id) ON DELETE CASCADE,
  kind        VARCHAR(30) NOT NULL, -- 'slack' | 'teams' | 'pagerduty'
  | 'email' | 'discord'
  name        VARCHAR(100) NOT NULL,
  config       JSONB NOT NULL,
  enabled      BOOLEAN NOT NULL DEFAULT TRUE,
  created_at   TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
ALTER TABLE alerts ADD COLUMN channel_ids UUID[];
```

Each channel implements a `NotificationChannel` trait:

```
#[async_trait]
pub trait NotificationChannel: Send + Sync {
  async fn send(&self, event: &AlertEvent) -> Result<()>;
  fn kind(&self) -> &str;
}
```

Implementations:

- **Slack** — Bot API (`chat.postMessage` with blocks); workspace + channel selector via OAuth in dashboard
- **Microsoft Teams** — incoming webhook + AdaptiveCard payload
- **PagerDuty** — Events API v2 (`trigger`, `acknowledge`, `resolve` lifecycle)
- **Email** — `lettre` crate, SMTP credentials per workspace
- **Discord** — webhook + embed

Alerts can target multiple channels via `alerts.channel_ids[]`.

9.2 APM exporters

Generic **MetricsExporter** trait alongside the existing Prometheus path:

- **Datadog** — **dogstatsd** UDP + Logs API
- **New Relic** — Insights Events API (**POST** **/v1/accounts/:id/events**)
- **Honeycomb** — already supported via OTel (V3-2); document the recipe, no new code

Config:

```
[exporters.datadog]
enabled = true
api_key = "..."
site    = "datadoghq.com"

[exporters.newrelic]
enabled = true
license_key = "..."
account_id = "..."
```

Exporters consume the existing metrics atomic gauges + **tracing** events — no changes to the event sources.

9.3 SLA reports

Background task, last day of each month, generates per-workspace SLA report:

- Uptime % (from **requests** table — successful proxied requests / total)
- p50, p95, p99 latency
- Error rate by provider
- Top 5 costs
- Cache savings
- Output: HTML email to all **admin** role members of the workspace, plus PDF artifact retrievable via **GET** **/admin/sla/:workspace/:month**

9.4 Rate-limit analytics page (**/analytics/rate-limits**)

Reads existing rate-limit hit events from **requests** (status_code = 429). Shows:

- Keys with most 429s in last 7d
- Peak usage windows (heatmap by hour of week)
- Recommendation: "key X would benefit from raising rpm from N to M"

Pure frontend — no new backend code.

9.5 Function-calling analytics page (**/analytics/tools**)

Uses **requests.tool_calls** from V5-0:

- Most-called tools per key/workspace

- Avg latency / cost per tool
- Tool error rate (tool returned error in subsequent request)

Pure frontend.

9.6 Files to create

- `src/notifications/mod.rs` — `NotificationChannel` trait + dispatch
- `src/notifications/slack.rs`, `teams.rs`, `pagerduty.rs`, `email.rs`, `discord.rs`
- `src/exporters/mod.rs` — `MetricsExporter` trait
- `src/exporters/datadog.rs`, `newrelic.rs`
- `src/handlers/admin/channels.rs` — CRUD for notification channels
- `src/handlers/admin/sla.rs`
- `src/analytics/sla.rs` — monthly background task
- `dashboard/src/app/(dashboard)/notifications/page.tsx`
- `dashboard/src/app/(dashboard)/analytics/rate-limits/page.tsx`
- `dashboard/src/app/(dashboard)/analytics/tools/page.tsx`

9.7 Test contract

File: `tests/v5/v5_6_notifications.rs`

```
async fn v5_6_slack_channel_sends_block_payload()
async fn v5_6_teams_channel_sends_adaptive_card()
async fn v5_6_pagerduty_trigger_acknowledge_resolve_lifecycle()
async fn v5_6_email_channel_sends_via_smtp()
async fn v5_6_discord_channel_sends_embed()
async fn v5_6_alert_dispatches_to_multiple_channels()
async fn v5_6_channel_disable_skips_send()
async fn v5_6_channel_failure_does_not_block_others()
```

File: `tests/v5/v5_6_apm_exporters.rs`

```
fn v5_6_datadog_exporter_emits_dogstatsd_format()
fn v5_6_newrelic_exporter_emits_insights_event_format()
async fn v5_6_exporter_disabled_when_config_missing()
async fn v5_6_exporter_failure_isolated_from_request_path()
```

9.8 Definition of done

```
cargo test v5_6
# External signal: test alert reaches a real Slack workspace + real
PagerDuty incident triggered + Datadog dashboard shows velox_* metrics
```


10. Phase V5-7: Governance & Safety

Goal: Give security teams the answers they need before greenlighting Velox.

10.1 Policy engine

Admin-configurable rules evaluated in the request pipeline, between RBAC and rate-limit gates.

```
-- migrations/0033_policies.sql
CREATE TABLE policies (
  id          UUID PRIMARY KEY,
  workspace_id UUID REFERENCES workspaces(id) ON DELETE CASCADE,
  name        VARCHAR(100) NOT NULL,
  condition    JSONB NOT NULL,      -- {"all":[{"model":"gpt-4o"},
{"tag.team":"free-tier"}]}
  action       JSONB NOT NULL,      -- {"reject":{"message":"..."}}
  priority     INTEGER NOT NULL,
  enabled      BOOLEAN NOT NULL DEFAULT TRUE,
  dry_run      BOOLEAN NOT NULL DEFAULT FALSE,
  created_at   TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE TABLE policy_events (
  id          UUID PRIMARY KEY,
  policy_id    UUID NOT NULL REFERENCES policies(id),
  request_id   UUID,
  matched      BOOLEAN NOT NULL,
  action_taken VARCHAR(20) NOT NULL,
  created_at   TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

Condition DSL (JSON, evaluated server-side — no eval, no scripting):

```
{
  "all": [
    { "model": "gpt-4o" },
    { "tag.team": "free-tier" }
  ],
  "any": [
    { "prompt_matches_pattern": "credit_card" },
    { "prompt_matches_pattern": "ssn" }
  ]
}
```

Actions:

- **reject** (return 403 with message)
- **downgrade** (override **model** or routing strategy)
- **route_to** (force provider id)
- **redact** (strip matched pattern from prompt before forwarding)
- **tag** (add a tag — useful for audit cohorts)

- `log_only` (matches but does nothing else; combine with `dry_run` for soft-launch)

Dry-run mode: matched events written to `policy_events` but action skipped, so admins can validate before enforcement.

10.2 Prompt injection detection

Optional V3-4 plugin shipped as part of V5-7. Uses:

- Pattern library (curated list of known jailbreaks)
- Optional small ONNX classifier (already linked via embedding model — no new heavy dep)

Sets `X-Velox-Injection-Score: 0.0-1.0` header. Combine with policy engine to act on it: `{"injection_score_gt": 0.7}` → reject.

10.3 Content moderation hook

Optional pre-flight call to:

- OpenAI Moderation API (free), or
- Local ONNX moderation model (configurable)

Behavior configurable: `block` | `flag` | `log`. Adds latency only when enabled.

10.4 Egress allowlist

Per workspace: `allowed_providers UUID[]` on `workspaces`. Router refuses any provider not in the list. Combined with V5-4 data residency = strong egress controls.

10.5 Dashboard

New `/governance` page:

- Policy list with priority ordering (drag-to-reorder)
- Visual rule builder (no JSON editing required) — condition tree + action selector
- Dry-run toggle per policy
- Audit drawer: last N events for any policy, with matched request snippet
- Templates: "Block PII to public providers", "Cost cap for non-prod", "Compliance mode for HIPAA workspace"

10.6 Files to create

- `src/policy/mod.rs` — `Policy` + `Condition` + `Action` types
- `src/policy/eval.rs` — condition evaluator
- `src/policy/builtin_patterns.rs` — curated regex set (pii, secrets, jailbreaks)
- `src/handlers/admin/policies.rs`
- `src/plugins/injection_detector.rs`
- `src/plugins/moderation.rs`
- `dashboard/src/app/(dashboard)/governance/page.tsx`
- `dashboard/src/components/PolicyBuilder.tsx`

10.7 Files to modify

- `src/gateway/pipeline.rs` — evaluate policies after RBAC, before rate-limit
- `src/state.rs` — add `policy_engine: Arc<PolicyEngine>` cache

10.8 Test contract

File: `tests/v5/v5_7_policy_engine.rs`

```
fn v5_7_condition_all_matches_when_all_true()
fn v5_7_condition_any_matches_when_one_true()
fn v5_7_pattern_match_detects_credit_card()
fn v5_7_pattern_match_detects_ssn()
async fn v5_7_reject_action_returns_403()
async fn v5_7_downgrade_action_overrides_model()
async fn v5_7_route_to_action_forces_provider()
async fn v5_7_redact_action_modifies_prompt_before_forward()
async fn v5_7_dry_run_logs_match_but_does_not_act()
async fn v5_7_priority_order_evaluated_correctly()
async fn v5_7_disabled_policy_skipped()
async fn v5_7_egress_allowlist_blocks_unlisted_provider()
async fn v5_7_injection_score_attached_when_plugin_enabled()
async fn v5_7_moderation_block_action_rejects_disallowed_content()
async fn v5_7_regression_no_policies_means_no_overhead()
```

10.9 Definition of done

```
cargo test v5_7
# External signal: ship 3 template policies; one of them catches a planted
PII test request in dry-run and emits an audit event
```

11. Phase V5-8: Performance & Reliability Proofs

Goal: When a buyer asks "how fast? how reliable?" you have numbers, they're public, and they were produced by CI.

This phase is mostly tests + infra. Zero new product features.

11.1 Load test suite

`benches/load/` — k6 scripts driven by GitHub Actions:

- `01-baseline.js` — 100 req/s for 5 min, chat completions, mixed models
- `02-spike.js` — 500 req/s burst from 50
- `03-sustained.js` — 1000 req/s for 5 min
- `04-dedup-amplification.js` — N identical concurrent requests; verify only one provider call
- `05-cache-cold-to-warm.js` — measure cache fill rate + hit ratio trajectory

CI workflow [.github/workflows/load.yml](#) runs these on a known instance type (e.g. AWS c6i.xlarge) nightly + on tagged releases. Results posted to S3 + rendered in [docs/benchmarks/](#).

11.2 Chaos test suite

[tests/v5/v5_8_chaos.rs](#) — uses [toxiproxy-rs](#) for failure injection:

- Provider connection cut mid-stream → SSE downstream gets clean error
- DB connection drops → in-flight requests fail fast, new requests get 503 until pool recovers
- Provider returns 5xx → failover triggers
- Cache layer panics → falls through to provider
- OTEL exporter dead → zero impact on request path
- Disk fills (90%) → graceful degradation, warning logged

11.3 Playwright E2E suite

[dashboard/e2e/](#) — one suite per dashboard page (Alerts, Keys, Providers, Prompts, Playground, Cost Simulator, Workspaces, Notifications, Governance). Each suite: create / edit / delete the canonical entity, verify API calls succeeded, verify error states render.

CI workflow [.github/workflows/e2e.yml](#) runs against [velox demo](#) instance.

11.4 Performance regression CI

Criterion benches (existing [benches/cache_bench.rs](#) + new ones) gate PRs:

- Cache lookup p95
- Cosine scan over 10k entries
- Request pipeline overhead (auth → cache → forward)
- Embeddings inference p95

Any >10% regression vs [main](#) baseline fails CI.

11.5 Public benchmark page

[docs/benchmarks/index.md](#) — auto-updated from CI:

- Latest run timestamp + Velox version
- Chart: req/s vs p95 latency at 100/500/1000 RPS
- Chart: cache hit ratio over time
- Side-by-side comparison: Velox vs LiteLLM at matched config (same providers, same cache settings)

11.6 Files to create

- [benches/load/*.js](#)
- [tests/v5/v5_8_chaos.rs](#)
- [dashboard/e2e/](#) — Playwright config + per-page specs
- [.github/workflows/load.yml](#), [e2e.yml](#), [bench-regression.yml](#)
- [docs/benchmarks/index.md](#) + auto-generated subpages

11.7 Test contract

File: `tests/v5/v5_8_chaos.rs`

```
async fn v5_8_chaos_provider_disconnect_mid_stream_emits_clean_error()
async fn v5_8_chaos_db_disconnect_returns_503_until_pool_recovers()
async fn v5_8_chaos_provider_5xx_triggers_failover()
async fn v5_8_chaos_cache_panic_falls_through_to_provider()
async fn v5_8_chaos_otel_exporter_dead_zero_impact()
```

11.8 Definition of done

```
cargo test v5_8
npx playwright test
# CI gate: load tests green, regression CI green
# External signal: docs/benchmarks/index.md committed with real numbers
```

12. Phase V5-9: Docs, Onboarding, and GTM Assets

Goal: Convert visitor → first successful request in under 5 minutes, and remove brand confusion.

This is mostly content + a small Next.js site. Zero new backend code.

12.1 License switch (ELv2)

- Replace `LICENSE` file (MIT → ELv2)
- Update `Cargo.toml license = "Elastic-2.0"`
- Update README + `docs/` license headers
- Update package metadata in `velox-python/` and `velox-node/`

12.2 Brand cleanup

- Replace every "Anthropic" mention in README, CHANGELOG, docs, dashboard footer
- Replace `ghcr.io/anthropics/velox` image path with new namespace
- Replace `support@anthropic.com` with new support email
- Update repo description on GitHub

12.3 Marketing landing (`velox.dev`)

Separate Next.js site at `marketing/` in repo (or separate repo `velox-www`):

- Hero: positioning statement, demo gif, "Run in 60 seconds" CTA
- Social proof: install command, GitHub star count, customer logos when present
- Comparison table: Velox vs LiteLLM/Portkey/Helicone/Cloudflare
- Features grid (cached, observable, governed, multi-tenant, self-hosted)
- Pricing: "Self-hosted: free under ELv2. Managed cloud: waitlist."

- Footer: docs, GitHub, Discord, status page, blog

Hosted on Vercel/Cloudflare Pages. Custom domain.

12.4 Comparison pages

Each is a long-form honest comparison:

- `/vs/litellm` — Velox is a product, LiteLLM is a library. When to pick which.
- `/vs/portkey` — Both products. Self-hosted vs SaaS tradeoffs.
- `/vs/helicone` — Velox includes routing + caching; Helicone is observability-only.
- `/vs/cloudflare-ai-gateway` — Edge cost vs feature depth.

Each page ends with a `velox import <competitor>` CTA.

12.5 Quickstart 2.0

- 60-second demo video embedded on landing + in dashboard onboarding
- Interactive in-dashboard tour via `react-joyride`:
 1. Create your first key
 2. Send a test request from the Playground
 3. View it in Requests
 4. Configure your first alert
 5. (Optional) Connect SSO
- Tour state persisted in `users.tour_completed_steps[]`

12.6 Sample apps repo (`velox-examples`)

Each in `examples/<app>/` with a README:

- `rag-chatbot` — Python: pgvector + `/v1/embeddings` + `/v1/chat/completions`, uses `velox-python`
- `tool-using-agent` — Node: function calling + V5-0 tool analytics
- `multi-provider-failover` — Demonstrates V3 failover behavior under provider outage
- `embeddings-search` — Semantic search over a corpus, shows cache hit ratio over time
- `compliance-mode-deployment` — Velox configured for HIPAA-ready posture (V5-4 + V5-7 settings)

12.7 Docs site

`docs/` migrates to Mintlify or Docusaurus, served at `docs.velox.dev` (or embedded at `/docs` in Velox itself for self-hosted users):

- API reference: auto-generated from V5-1 OpenAPI
- Guides: deployment, RBAC, SSO, compliance, governance, cost attribution
- Tutorials: every sample app
- Migration guides: from LiteLLM/Portkey/Helicone

12.8 Community

- Discord server with channels: `#announcements`, `#general`, `#self-hosting`, `#feature-requests`, `#showcase`, `#contributors`
- GitHub Discussions enabled and triaged weekly
- CONTRIBUTING.md (was Anthropic-branded — needs rewrite)
- Code of Conduct
- Issue templates: bug, feature, question, security

12.9 Status page

`status.velox.dev` — Statuspage.io (free tier) or self-hosted Cachet. Subscribe-by-email, RSS, Slack integration. Used for managed-cloud beta (V6) but launched now to establish trust.

12.10 Blog launch sequence (5 posts over 4 weeks)

1. *Why I built Velox in Rust* — founder story
2. *Semantic cache deep dive: how a 384-dim vector saved \$X* — technical
3. *From LiteLLM to Velox in 10 minutes* — migration
4. *RBAC, audit logs, and SSO: the boring parts that matter* — enterprise readiness
5. *Cost attribution by team — without the spreadsheet* — finance

Each post syndicated to dev.to, Hacker News, `/r/MachineLearning`, `/r/selfhosted`, AI Engineer Slack/Discord communities.

12.11 Files to create

- `LICENSE` (replace MIT with ELv2)
- `marketing/` — Next.js landing site
- `examples/rag-chatbot/`, `tool-using-agent/`, etc.
- `docs/vs/litellm.md`, `portkey.md`, `helicone.md`, `cloudflare-ai-gateway.md`
- `dashboard/src/components/OnboardingTour.tsx`
- `.github/ISSUE_TEMPLATE/*.yml`
- `CONTRIBUTING.md`, `CODE_OF_CONDUCT.md`
- `blog/2026-06-01-why-rust.md` ... 5 posts

12.12 Files to modify

- `README.md` — full rewrite: new license, new brand, new positioning, link to landing
- `CHANGELOG.md` — V5 entries
- Every `docs/*.md` — strip "Anthropic" mentions
- `dashboard/src/app/layout.tsx` — footer brand
- `Cargo.toml` — package metadata, license field, authors

12.13 Definition of done

```
# Self-checks
grep -ri "anthropic" --include="*.md" --include="*.rs" --include="*.toml" \
  --include="*.tsx" --include="*.ts" .    # → no matches
```

```
grep -i "MIT" LICENSE # → no match (now ELv2)
cargo test # still green
```

External signals (all must be live before declaring V5 launched):

- velox.dev landing page live
- docs.velox.dev / [/docs](#) content complete
- Discord server live with launch announcement
- status.velox.dev live with [Velox Self-Hosted Build Pipeline](#) component
- ELv2 license switch committed
- 5 sample apps run end-to-end
- HN launch post drafted

13. What V5 Explicitly Does NOT Include

Item	Why deferred
Managed cloud (control plane, signup, billing)	V6 scope — too much surface for parallel work; opens with waitlist at V5-9 launch
Async batch API (/v1/batches)	Architectural; needs job queue. VELOX_FUTURE §3
Fine-tuning proxy	Async + large files, thin demand signal
Prompt compression as default behavior	Violates transparent-proxy contract; opt-in only via plugin
LLM-as-judge / evaluation harness	Different product category
Native trace viewer in dashboard	Jaeger / Grafana Tempo already excellent — OTel from V3-2
Horizontal semantic cache sync	V4-9 Qdrant already covers it
Mobile apps	No demand signal
Visual policy editor with NL→DSL conversion	V6 candidate after policy engine matures
Customer-facing usage portal	Managed cloud (V6) feature
AWS Marketplace AMI	V6 — deploy options shipped in V5-2 are sufficient

14. Migration Plan

Migration	Phase	Description
0001–0024	V1/V2/V3/V4	Existing schema
0025–0026	(pricing fixes, current)	Already on disk

Migration	Phase	Description
0027	V5-0	requests.tool_calls, requests.endpoint; model_pricing modality columns
0028	V5-3	identity_providers, identities, scim_tokens
0029	V5-4	requests.audit_hash, previous_hash, chain_sequence; audit_events
0030	V5-4	workspaces.region, allowed_regions; providers.region
0031	V5-5	requests.tags; daily_costs.forecast_*
0032	V5-6	notification_channels; alerts.channel_ids
0033	V5-7	policies, policy_events; workspaces.allowed_providers

SQLite migrations: maintained in `migrations/sqlite/` per V2-1 convention. Every PostgreSQL migration must have a matching SQLite migration before phase completion.

Rule (CLAUDE.md): never modify existing migrations. Each change is a new file.

15. Dependency Plan

V5-0

```
# No new heavy deps; existing request + multer (for multipart) cover audio uploads
multer = "3" # if not already present transitively
```

V5-1

```
utoipa          = { version = "5", features = ["axum_extras", "uuid", "chrono", "decimal"] }
utoipa-swagger-ui = { version = "8", features = ["axum"] }
clap             = { version = "4", features = ["derive", "env"] } #
confirm not already pinned
```

V5-2

```
flate2 = "1" # backup tar.gz
tar     = "0.4" # backup archive
```

V5-3

```
openidconnect = "3"
samael        = "0.0.18"    # SAML – confirm latest at implementation time
```

V5-4

```
hashicorp_vault    = "2"                # or custom thin client
aws-sdk-secretsmanager = "1"
google-cloud-secretmanager = "0.5"
```

V5-5

```
printpdf = "0.7"    # invoice PDF
```

V5-6

```
lettre = { version = "0.11", features = ["smtp-transport", "tokio1-native-tls"] }
# Slack, Teams, Discord, PagerDuty use existing request – no new deps
# Datadog uses cadence (dogstatsd) – light
cadence = "1"
```

V5-7

```
# No new heavy deps – policy engine uses serde_json + regex (both present)
```

V5-8

```
[dev-dependencies]
toxiproxy_rust = "0.2"    # chaos testing
```

V5-9

```
# No Rust deps – content + Next.js site only
```

16. V5 Phase Status Tracker

Phase	Type	Description	Status	Migration
V5-0	Backend	API Surface Expansion (embeddings, images, audio, models, tools)	 Not started	0027
V5-1	Backend + SDKs	SDKs, CLI, OpenAPI	 Not started	—
V5-2	Backend + Infra	Deployment & Migration Tooling	 Not started	—
V5-3a	Backend + Frontend	OIDC + SCIM	 Not started	0028
V5-3b	Backend + Frontend	SAML 2.0	 Not started	0028 (shared)
V5-4	Backend	Compliance & Audit Hardening	 Not started	0029, 0030
V5-5	Backend + Frontend	Cost Attribution & FinOps	 Not started	0031
V5-6	Backend + Frontend	Notifications & APM Integrations	 Not started	0032
V5-7	Backend + Frontend	Governance & Safety	 Not started	0033
V5-8	Tests + CI	Performance & Reliability Proofs	 Not started	—
V5-9	Content + Frontend	Docs, Onboarding, GTM Assets	 Not started	—

17. Locked Decisions Record

Decisions made before V5-0 begins. Update DECISIONS.md with these.

#	Topic	Decision	Reasoning
L1	License	Switch MIT → Elastic License v2	Self-host stays free; blocks AWS-style hosted resale; same path Elastic/MongoDB/Redis took
L2	Brand	Independent — Velox by Farzad Alizadeh , remove all "Anthropic" references	Solo developer product, not Anthropic-owned
L3	Managed cloud	Deferred to V6	Solo dev cannot ship self-host V5 + managed cloud in parallel; managed needs ≥20 self-host customers first

#	Topic	Decision	Reasoning
L4	SSO order	OIDC first (V5-3a), SAML second (V5-3b)	OIDC covers ~80% of buyers per week of work
L5	CLI shape	Single velox binary with subcommands via clap	Standard ergonomics (kubect!/docker/gh); one PATH entry
L6	OpenAPI tool	utoipa + utoipa-swagger-ui	Compile-time generation prevents spec drift
L7	Sample-app stack	Python primary, Node secondary	Python is dominant in AI/RAG today; covers more buyers
L8	Docs platform	Mintlify (or Docusaurus fallback)	Mintlify gives fastest setup; Docusaurus if self-host is required
L9	Postgres in Helm	External Postgres only	Production should never run Postgres in the same Pod as the app

18. Session Start Ritual for V5 Work

```
# 1. Confirm V4 (and all earlier) is still green
cargo test 2>&1 | tail -20

# 2. Check V5 phase status (this file, §16)

# 3. Run the specific phase tests if work is in progress
cargo test v5_0 # (or v5_1, v5_3, etc.)

# 4. Confirm decisions in §17 are reflected in the codebase
grep -ri "anthropic" --include="*.md" --include="*.rs" . # should taper
to 0 by V5-9

# 5. Tell the user: "We are on Phase V5-X. Sub-phase: Y. Ready to
continue."
```

Do NOT write any code until you have done all 5 steps.

Glossary

Term	Meaning in V5
Adoption blocker	A gap that causes a real prospect to say "we cannot adopt Velox until this is fixed"
External signal	A verifiable artifact (download, working demo, real Slack message) that proves a phase removed its blocker

Term	Meaning in V5
Self-serve	A developer evaluating Velox alone, without sales contact
Enterprise deal	An organization where IT, security, finance, and procurement each review separately
ELv2	Elastic License v2 — source-available, free for end-users and self-hosters, restricted for hosted-resale
DSAR	Data Subject Access Request (GDPR Article 15)
RTBF	Right To Be Forgotten (GDPR Article 17)

Created: 2026-05-24 — based on V4 complete (all 10 phases) and market-readiness gap analysis
Locked decisions: §17 — do not change without explicit conversation + DECISIONS.md update
Update the Phase Status Tracker (§16) at the end of every session.